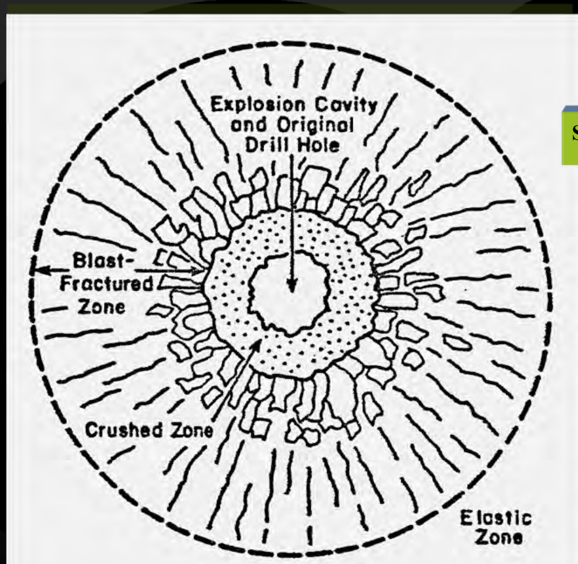
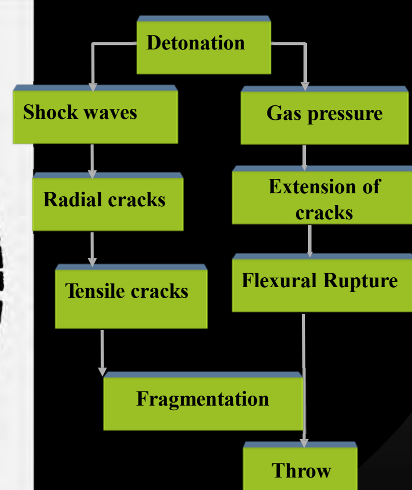
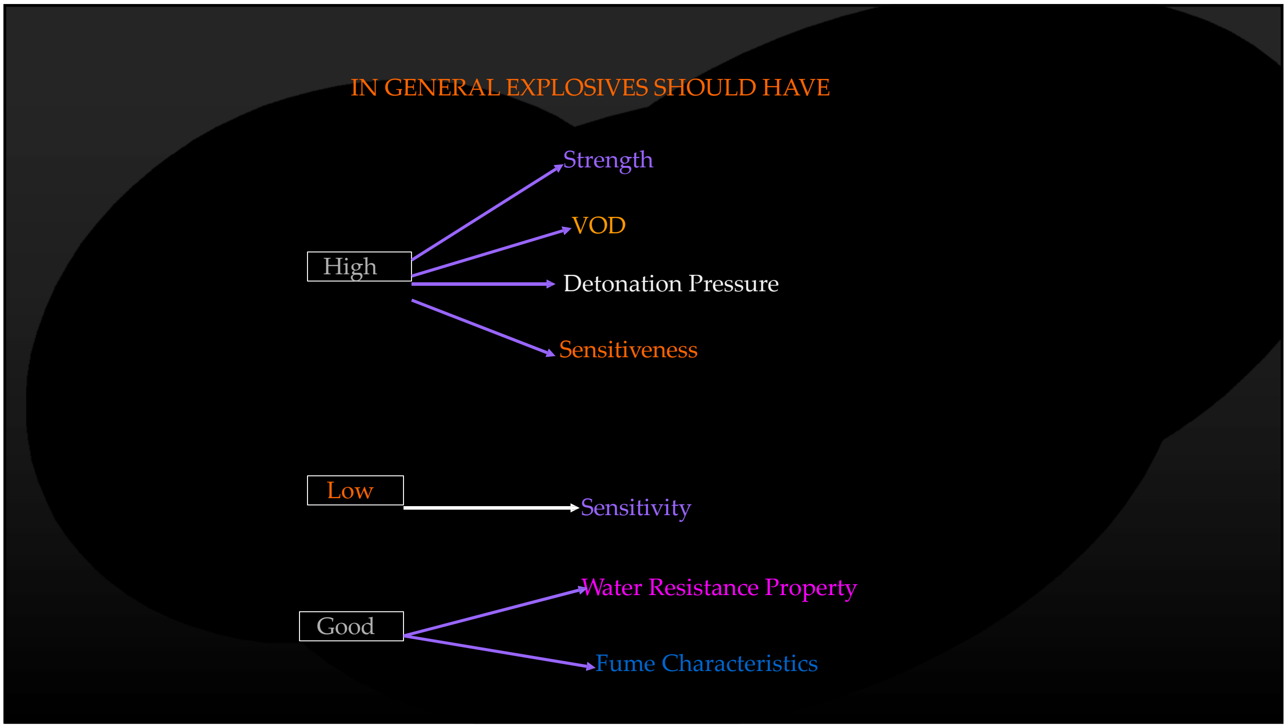
Blasting Techniques

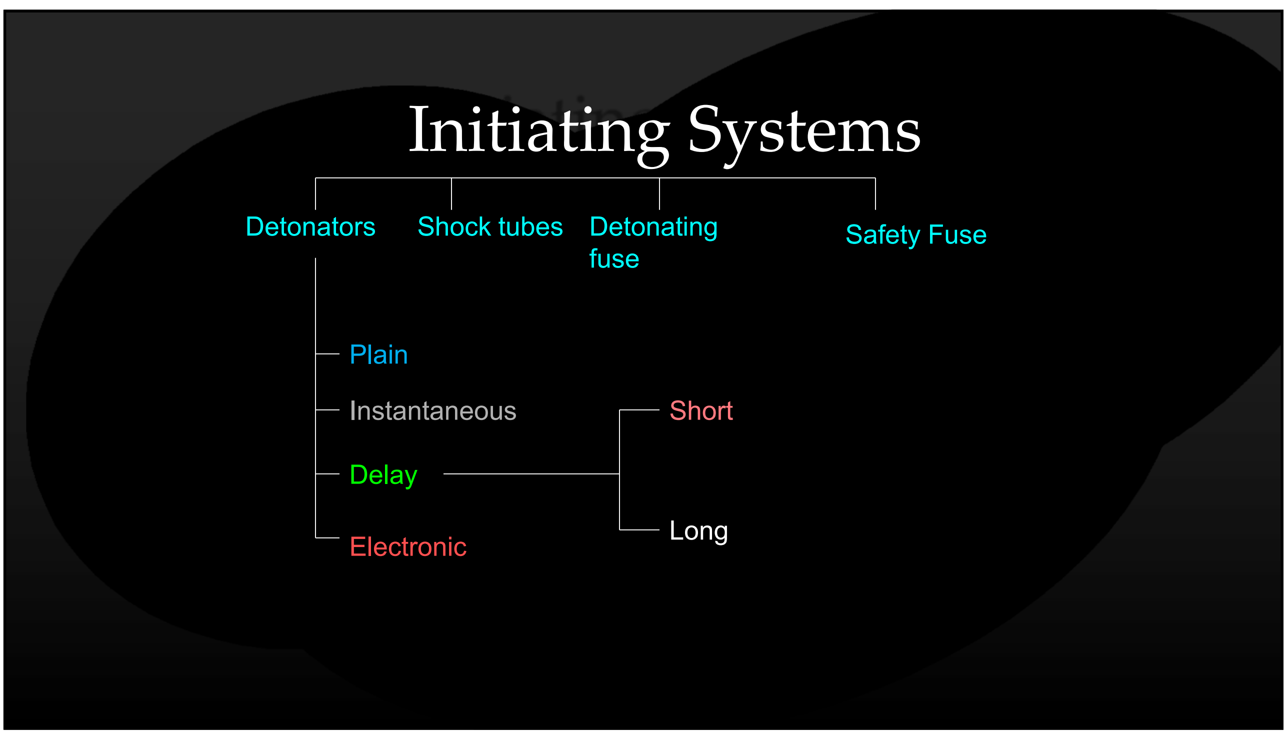
The Mechanics of Blasting



Fracturing and deformation around an explosion




IN GENERAL EXPLOSIVES SHOULD HAVE
High
Strength
VOD
Detonation Pressure
Sensitiveness
Low
Sensitivity
Good
Water Resistance Property
Fume Characteristics


Initiating Systems
Detonators
Shock tubes
Detonating fuse
Safety Fuse
Plain
Instantaneous
Delay
Electronic
Short
Long

Electric Delay Detonator

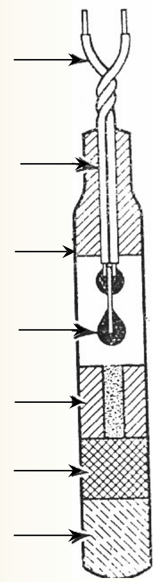
DELAY ELEMENT

(1) Composition:

Antimony Powder &
Potassium Permanganate

(2) Properties:

Gasless &
Fixed Burning Rate



Scattering in Delay Time

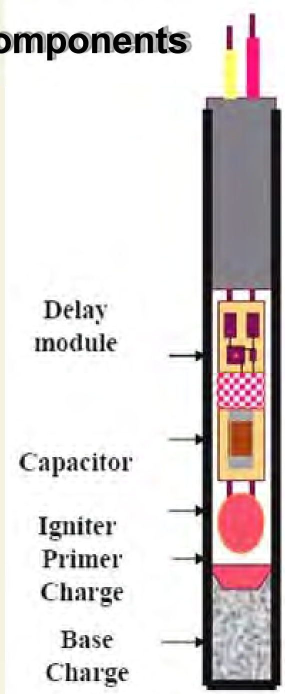
Delay no.	Nominal declared delay interval (ms)	Statistical analysis of measured delay timing				No. of Detonators outside the nominal declared range
		Min. (ms)	Max. (ms)	Average (ms)	Std. Deviation (ms)	
0	5-8	4.0	8.2	5.63	0.82	26
1	25 ±10	26	33.2	29.75	1.92	Nil
2	50 ±10	50.8	68.8	56.88	4.00	17
3	75 ±10	75.2	93.6	80.95	3.90	12
4	100 ±10	100.8	120.0	113.17	3.78	78
5	125 ±10	128	154.0	137.38	4.27	74
6	150 ±10	152	188.0	166.65	6.83	76

PROBLEMS DUE TO OVERLAPPING OF DELAYS

- Increased ground vibration and air over pressure due to increase in the charge per delay
- Increased chances of strata instability
- Poor rock fragmentation with large amount of over size pieces.
- Increased need for secondary blasting
- Explosive flame may ignite the possible inflammable mixture of coal dust or methane – air mixture

Electronic Detonator - components

- Detonator Unit-
  - Fully Programmable,
  - Microchip, Capacitor, Oscillator Circuit, NV Memory
  - Change in Delay Module
  - Protection against Radio frequency, Stray current
- Logger Unit-
  - Communication device between detonator and Blaster unit
- Blaster Unit –
  - Firing device



ROCK BLASTING

Desirable Effects

- Breaking the rock mass into desired shapes & sizes.
- Displacing the broken rock
- High Production/ Pull

Unwanted Effects

- Ground Vibration
- Air-blast
- Flyrock
- Over-Breakage
- Back break
- Damage to Rock Mass
- Cost escalation
- Increased cycle time

BLAST EFFICIENCY PARAMETERS

- Productivity
  - Powder factor, kg/m^3
  - Specific drilling, m/m^3
  - Fragmentation, m^2/t
  - Ovebreak/ underbreak, m
  - Profile of muck, (^{o})
- B) Safety
  - Ground vibration, mm/s
  - Noise, dB (A)
  - Fly rock, m
  - Air overpressure, dB(L)

PARAMETERS INFLUENCING BLAST RESULTS

- **Non-controllable**
  - Rock properties
  - Strength
  - Joint properties
  - P-wave velocity - Impedance matching
- **Controllable**
  - Explosive properties
  - Charge density (kg/m)
  - VOD
  - Gas ratio
  - Coupling
  - Water resistance
- **Blast design parameters**
  - Hole diameter
  - Charge per unit volume of in-situ rock
  - Hole depth
  - Sequence and type of initiation

PATTERN OF HOLES

In drivage work placement of holes properly, while designing a *pattern of holes*, is of prime importance due to following reasons:

- To obtain a desired shape, size, orientation and gradient of the mine openings and tunnels.
- To achieve an accurate contour of the tunnels or mine openings, with least over break and smooth surface of the floor and face.
- For compact heaping of the blasted muck after blasting at the working face.

The following two techniques are available for rock excavation

1. Mechanized-cut kerf

2. Blasting off the solid

1. Mechanized-cut kerf

In this technique use of rock cutting machines is made to provide a cut/kerf/cavity, which can be put at the bottom, middle, top or at any other desirable position of a mine opening or tunnel face.

This acts as an initial free face towards which blasting of the holes drilled in the face is directed.

This free face reduces amount of drilling and explosives considerably; but cutting the kerf is practicable in soft and medium hard rocks such as coal, salt, potash etc. Hence, its use is usually restricted to coal mines only.

2. Blasting off the solid

This is a universal technique applicable for any type of strata, tunnel or mine. A particular pattern is followed to position the holes.

Types of pattern of holes mainly differ in the arrangement of breaking in holes (known as cut holes i.e. the holes, which are used to create an initial free face, towards which the blast is subsequently directed), easers, trimmers and line (side) holes.

Broadly, these patterns can be classified as:
1. Parallel hole cuts
2. Angled cuts.

Parallel hole cuts

Parallel hole cuts If braking in holes are put at right angle, or parallel to the direction of the working face, these types of pattern of holes are known as parallel hole cuts.

Burn cut, cylindrical cut and cormorant cut, fall in this category.

A cluster of parallel shot holes, known as ‘cut’; is drilled almost parallel to the intended direction of face to blastoff a cavity in the center of the heading.

Some of the holes are charged while others are kept empty. The shock waves when reflected at these empty holes, the rock is shattered and subsequently blown out by the escaping gases.

There is specific geometrical relationship between diameter of empty holes and spacing between the empty and charged holes, in a given rock, which gives essential condition of breakage.

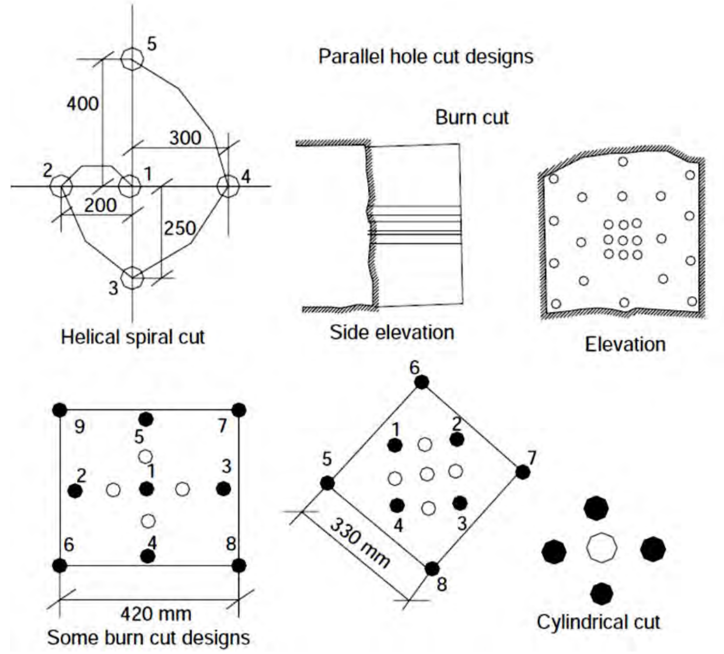
With spacing less than 1.5 times the diameter of empty hole, the blasting is expected to be clean while with higher spacing the breakage would tend to be uneven. With spacing in excess of twice the diameter of empty hole, there is a likely hood of plastic deformation and burning of rock.

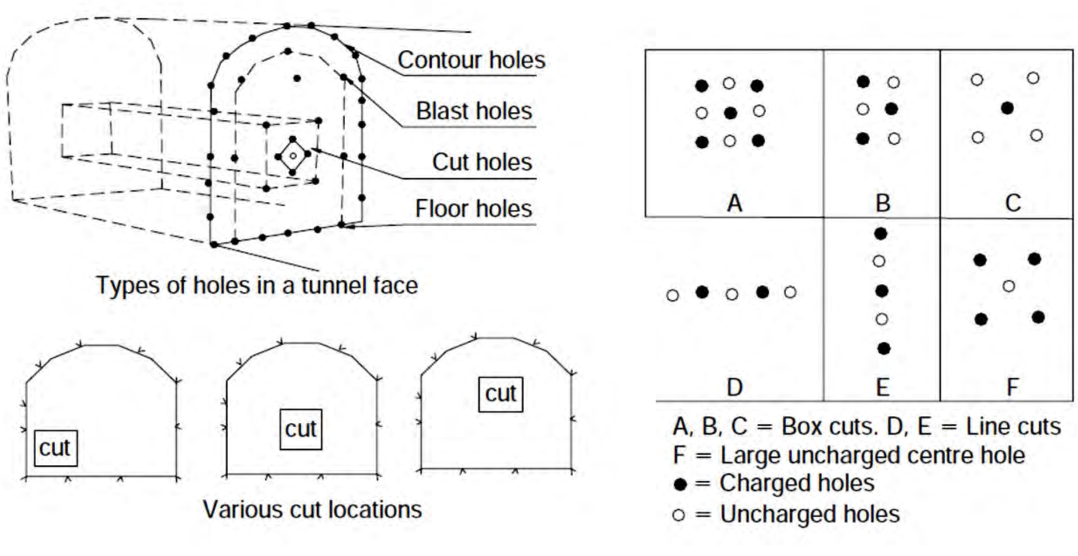
For achieving the greater advance, it is therefore, necessary to increase the diameter of empty hole, which, in turn gives better scope for increasing spacing between the holes without jeopardizing the condition of free breakage.

Even with proper burden if the charge concentration in the hole is too high, a miss-function of cut may result due to rock impact and sintering, preventing generation of the planned free face.

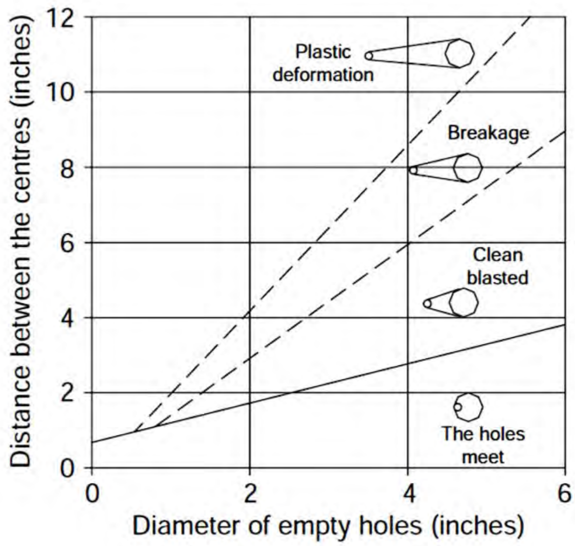
There are variations in shot hole patterns employing large diameter empty holes or providing empty space by not blasting few shot holes.

These patterns are suitable for hard, brittle and homogeneous rocks. These patterns are also known as fragmentation or fracture cuts.





Pattern of holes – parallel hole cuts



Blasting results in relation to empty hole's/holes' diameters and its distance from the charged blast-hole

The cut:

For good results consideration for these parameters must be given:

- Diameter of large empty hole/holes
- The burden
- The charge concentration.
- Drilling precision and charging skill.

In parallel hole cut, the advance of the round is restricted by the diameter of the empty hole and the deviation of the charge holes. Since drifting will be quite expensive if the advance is much less than 95% of the drilled hole depth, the average advance (I) can reach 95% of the blasthole depth (H).

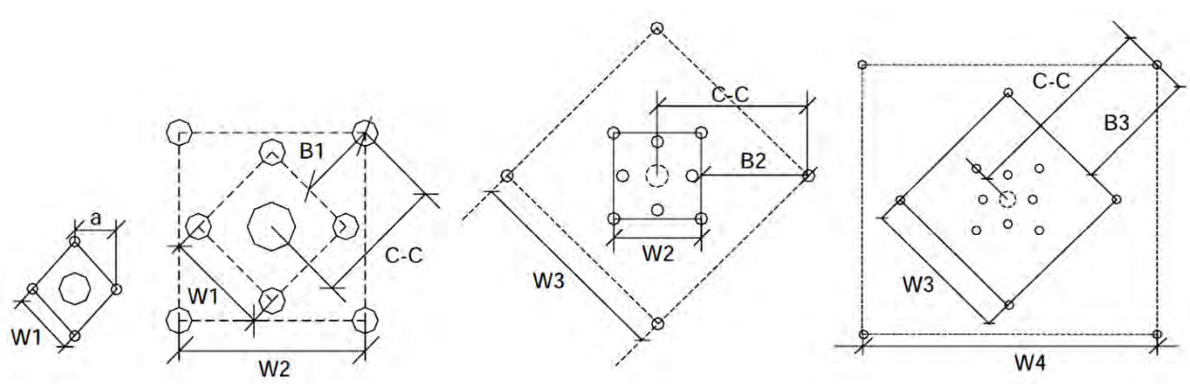
$$I = 0.95H$$

The breakage mechanism is greatly dependent upon the type of explosive, the rock properties and the distance between the empty hole and the charged blastholes.

Larger the empty hole diameter better it is, and deeper could be the depth of round. Slight deviation in drilling can jeopardize the complete blast.

Too large a burden will cause breakage only, or plastic deformation in the cut area.

But right burden will result in a clean blast up to the planned length (depth) of round, and thereby, better advance per round.



For best results it is taken as 1.5 D2; whereas D2 is the diameter of empty hole. Where several empty holes are used, a fictitious diameter, D, could be calculated using the following relation

$$D = D_2 \sqrt{n}$$

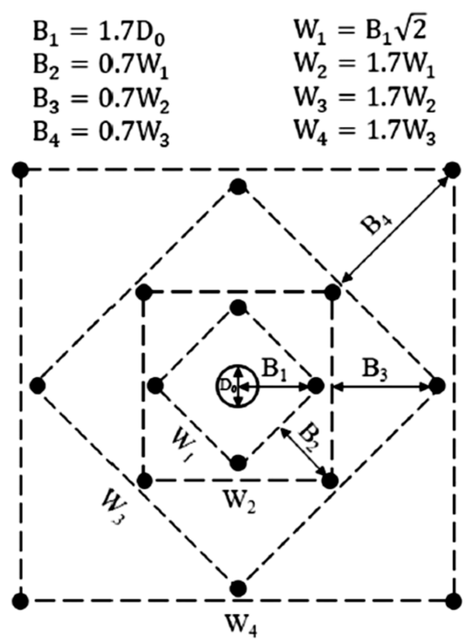
D2 is diameter of empty hole and n = number of holes.

In order to calculate the first square in the cut area, this D is used.

$$B_1 = 1.5\ D_2$$

B1 is the center-to-center distance between the large hole and the shot hole.

In case of several large holes; $B_1 = 1.5D$



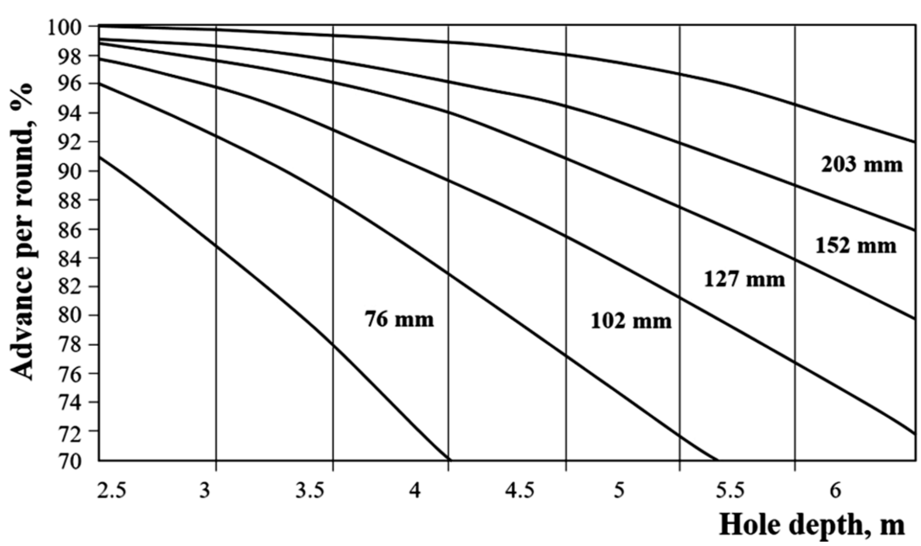
A typical four-section large hole cut design

For the overall blasting procedure, determining the burden of cut holes is crucial.

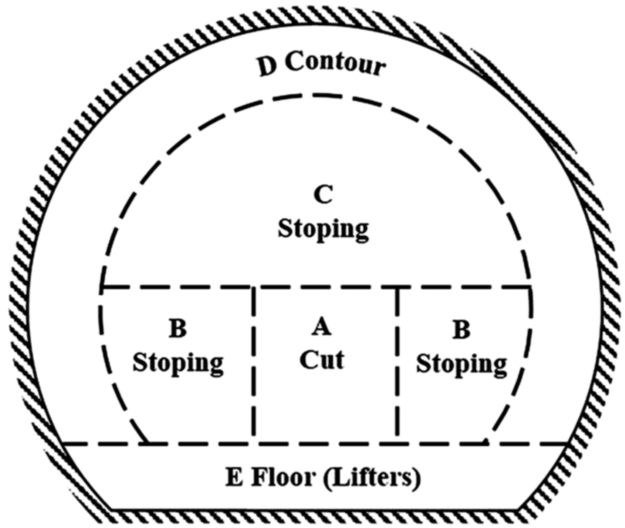
For an effective breakage process, the distance between the empty hole and the blasthole in the first quadrangle B1 should not exceed 1.7 times the diameter of the empty hole D_0.

For burdens larger than $2D_0$, the break angle is too small and a plastic deformation of rock between the blastholes is produced.

If the rock burden is under D_0, cut failure will occur. For this reason it is recommended that the burdens to becalculated as $B1 = 1.5D_0$.



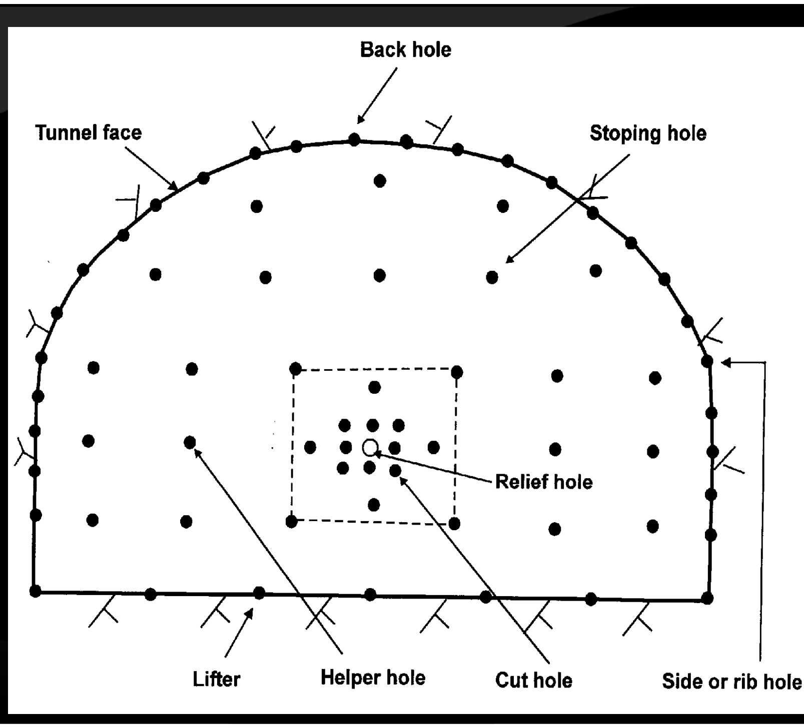
The relationship between hole depth and advance for different empty hole diameters



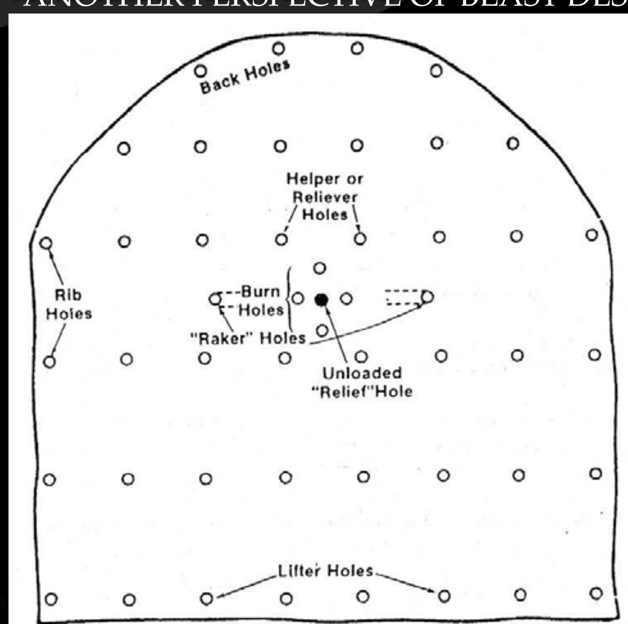
Zones in tunnel blasting

Part of the round	Burden, B (m)	Spacing, E (m)	Height of bottom charge, h_d (m)	Charge concentration		Stemming (m)
				Bottom, q_b (kg/m)	Column, q_c (kg/m)	
Stoping holes						
Upward and horizontally	B	1.1 × B	H/3	q_b	0.5 × q_d	0.5 × B
Downwards	B	1.2 × B	H/3	q_b	0.5 × q_d	0.5 × B
Roof holes	0.9 × B	1.1 × B	H/6	q_b	0.3 × q_d	0.5 × B
Wall holes	0.9 × B	1.1 × B	H/6	q_b	0.4 × q_d	0.5 × B
Floor holes	B	1.1 × B	H/3	q_b	1 × q_d	0.2 × B

Drilling and charging geometry for parallel hole cut



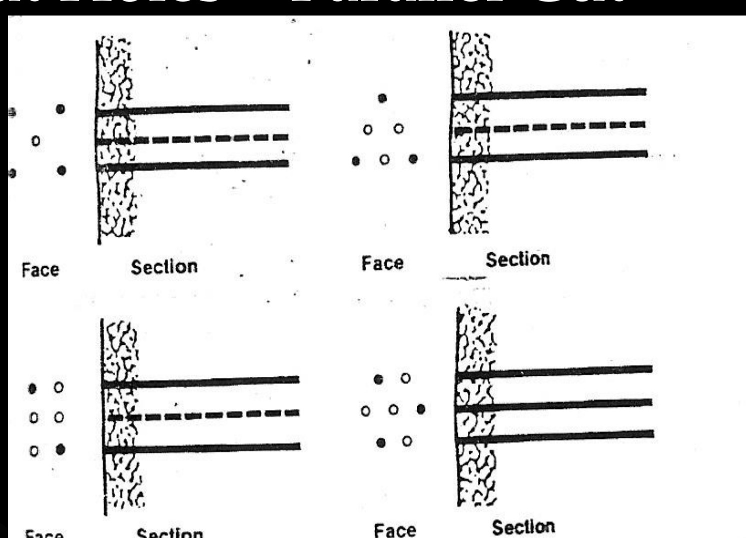
ANOTHER PERSPECTIVE OF BLAST DESIGN



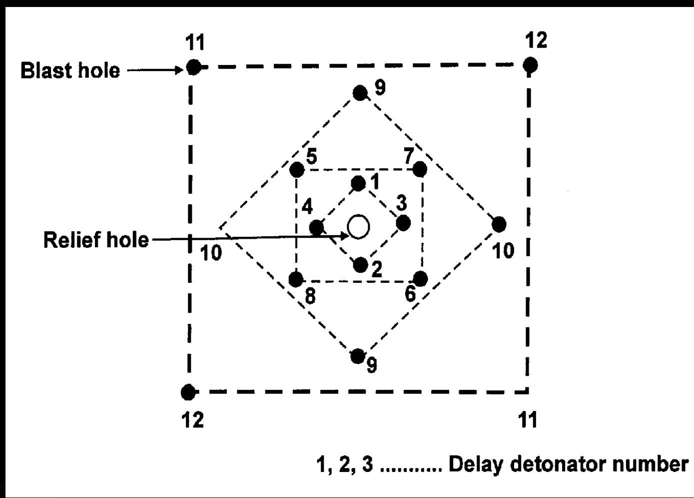
Cut Area

- Parallel Cut/ Burn Cut
- Wedge Cut/ Convergent/baby cut holes

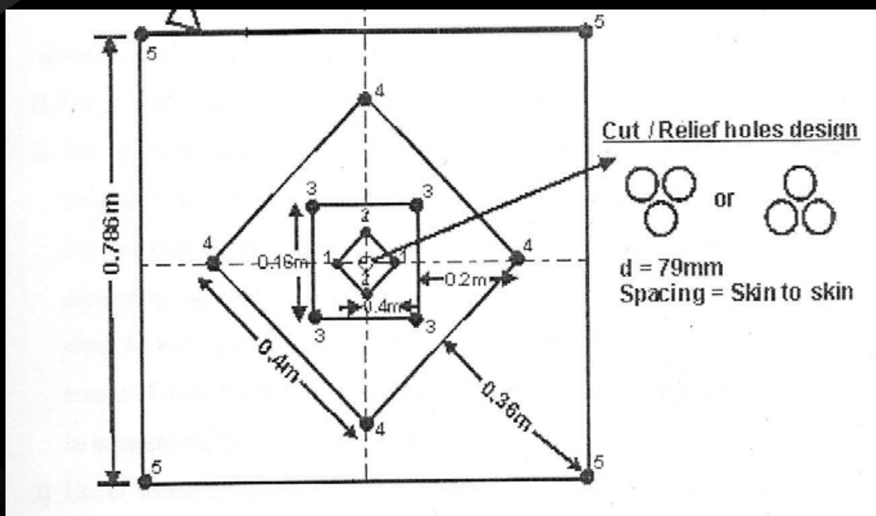
Cut Holes – Parallel Cut



Parallel Cut



Relief Holes



Essentials in Parallel cut

- Atleast 15 per cent of the cut area should be the relief hole area.
- Void requirement should be less in hard, brittle and unfractured formations and more in weak and fissured ones.
- First hole should pull about 20 per cent of the round depth
- Use of half-second delays in the cut holes.
- Drilling accuracy plays a major role in the success

The main difference between tunnel blasting and bench blasting is that tunnel blasting is done towards one free surface while bench blasting is done towards two or more free surface.

Various drilling patterns have been developed for blasting solid rock faces, such as:

i. wedge cut or V cut
ii. pyramid or diamond cut
iii. drag cut
iv. fan cut
v. burn cut

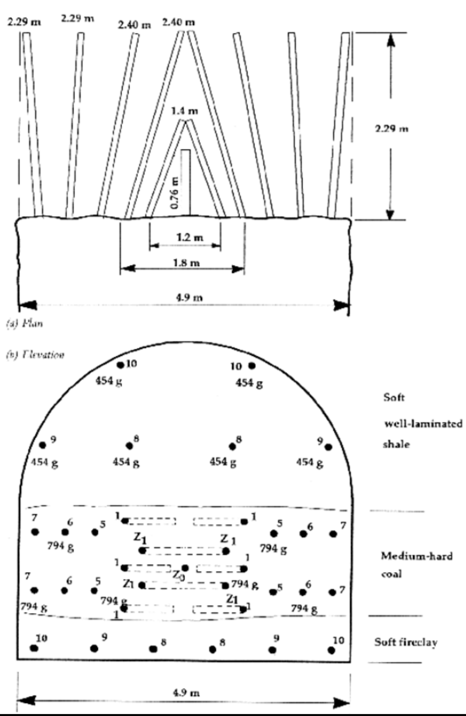
Wedge cut

Blasthole are drilled at an angle to the face in a uniform wedge formation so that the axis of symmetry is at the centre line of the face.

The cut displaces a wedge of rock out of the face in the initial blast and this wedge is widened to the full width of the drift in subsequent blasts, each blast being fired with detonators of suitable delay time.

The apex angle is as near as possible to 60^{o}

This type of cut is particularly suited to large size drifts, which have well laminated or fissured rocks. Hole placement should be carefully preplanned and the alignment of each hole should be accurately drilled.



Wedge cut

Wedge cut/ V cut: In ‘V’ cut two holes are drilled at a horizon (1-1.5 m above the floor) almost at the center of the face (width-wise). Ideally both holes should meet at their apex, but in practice it is difficult to achieve and thereby instead of ‘V’ a wedge is resulted, and hence the name ‘wedge cut’.

The angles of subsequent holes drilled at the same horizon are increased in such a way that the round holes (at the sides) are at 90-95° to the face. In harder strata (formations) a double or triple ‘V or Wedge’ can be drilled. While blasting, ‘V’ holes are given the initial delay and to the subsequent holes delays are given in an increasing order. Thus, a wedge of rock that is pulled out first is ultimately converted into a slot/kerf/slit that has been generated by way of blasting. The rest of the holes of the pattern are drilled parallel to the axis of the drive/tunnel and blasted in a sequential order using delay detonators and taking advantage of the initial free face created. This pattern is suitable for medium hard to hard strata.

Wedge Cut

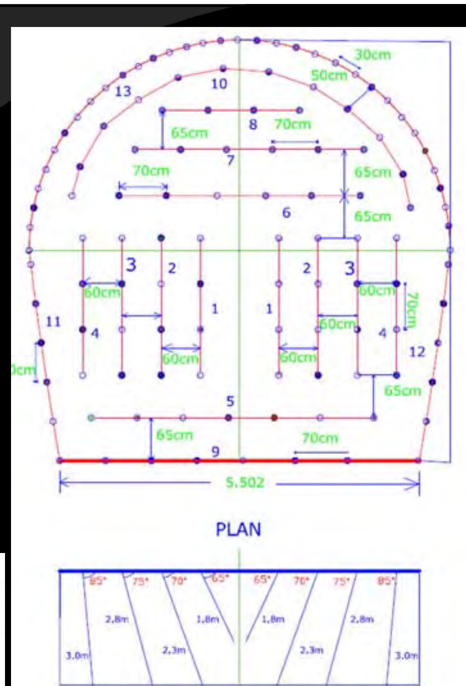


Fig:1 Blast design for tunnel excavation

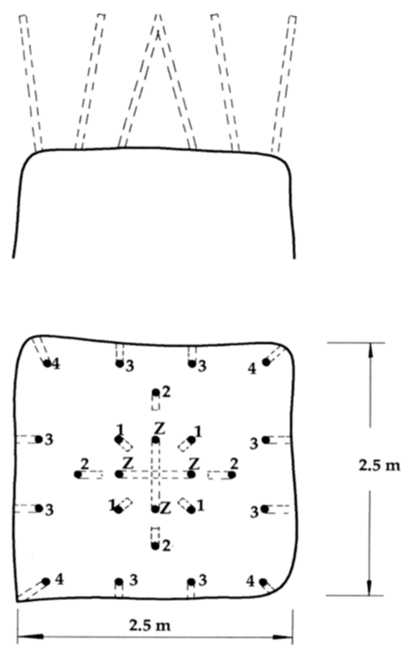
Wedge Cut Vs Parallel cut

- Wedge cut is suitable for tunnel large cross-section area
- The cut depth is always lower in a convergent cut than in a parallel cut
- The advance per round in a parallel cut is designed mainly on the basis of the relief hole size, whereas, the same in a convergent cut is decided by the tunnel size
- The cut holes in a parallel cut need to be placed very close to each other. Hence, there is a high possibility of joining the holes at the end, if the deviation is large
- The muck is thrown to a lesser distance in a parallel cut
- Mechanised drilling is essential for large diameter relief hole in a parallel cut
- More holes in Parallel cut.
- Higher drilling accuracy in a wedge cut holes

Pyramid or diamond cut

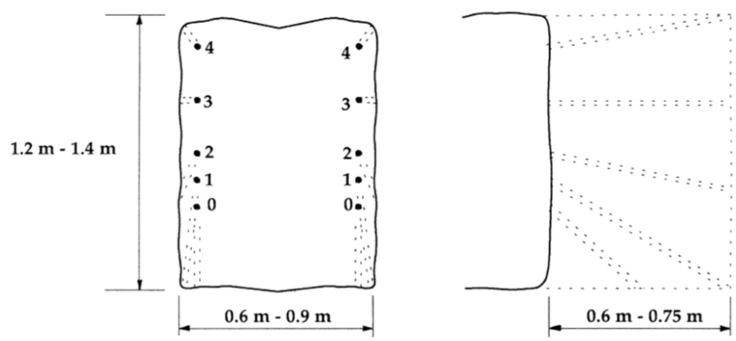
The pyramid or diamond cut is a variation of the wedge cut where the blastholes for the initial cavity may have a line of symmetry along horizontal axis as well as the vertical axis.

This pattern can be drilled for any drivage work-horizontal, up or downward. A cluster of holes (having 4-6 holes) is drilled in the center of face directing towards a common apex so that a pyramid is formed after blasting. The angles of subsequent holes drilled around the cut holes are increased in such a way that the round holes (at the sides) are 90–95° to the face. This pattern is popular during shaft sinking and raising operations and suitable for any type of strata.



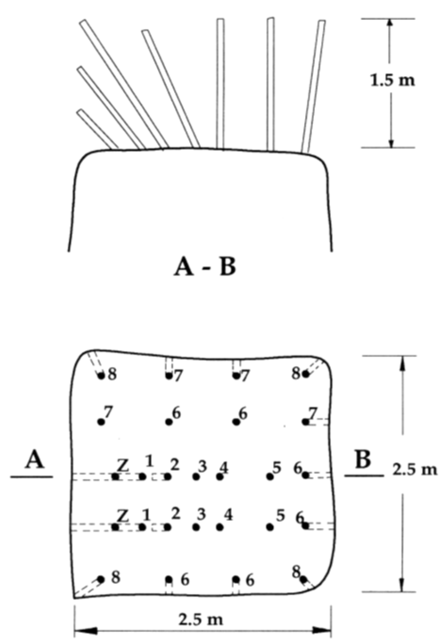
Drag cut

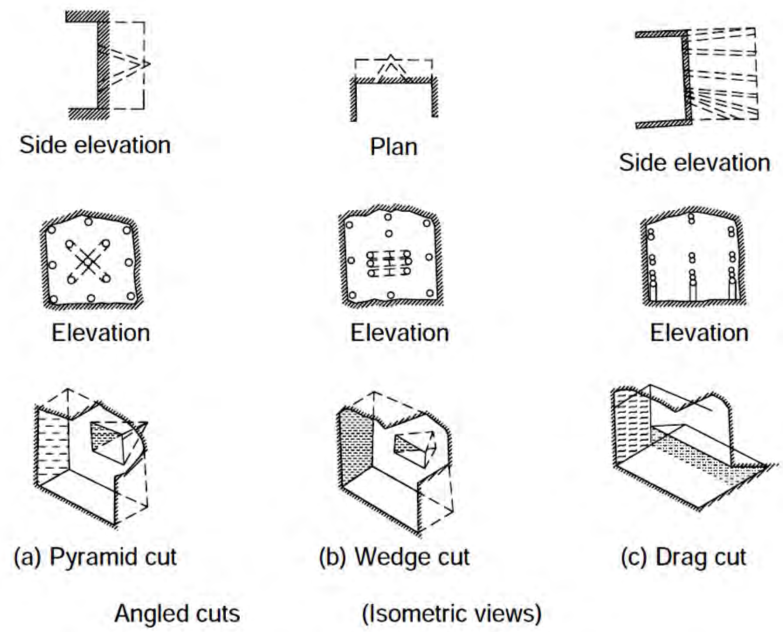
The drag cut is particularly suitable in small sectional drifts where a pull of up to 1 m is very useful. In this pattern, holes are drilled like a fan cut pattern but in a vertical plane. When these breaking in holes i.e. cut holes are blasted, an undercut (slot/ditch) at the face is created. The drilling and blasting for the rest of the holes are directed towards this cavity. This pattern is suitable for soft to medium hard strata.



Fan cut

The fan cut is one-half of a wedge cut and is applicable mainly where only one machine is employed in a narrow drive. Generally the depth of pull obtainable is limited to 1.5 m. In this pattern holes are drilled in a fan like fashion. Holes are drilled at a horizon, 1-1.5 m above the floor, setting them at different angles in an increasing order, starting from one side of the face, so that last hole (at the other side of the face) is at 90–95°. This pattern is suitable for soft to medium hard strata.





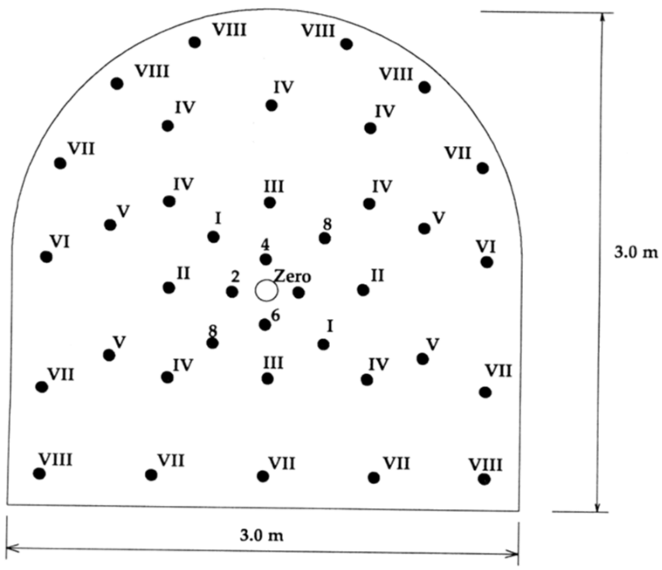
Burn cut

A series of parallel holes are drilled closely spaced at right angles to the face. One hole or more at the centre of the face are uncharged. This is called the burn cut.

The uncharged holes are often of larger diameter than the charged holes and form zones of weakness that assist the adjacent charged holes in breaking out the ground.

Since all holes are at right angles to the face, hole placement and alignment are easier than in other types of cuts. The burn cut is particularly suitable for use in massive rock such as granite, basalt etc.

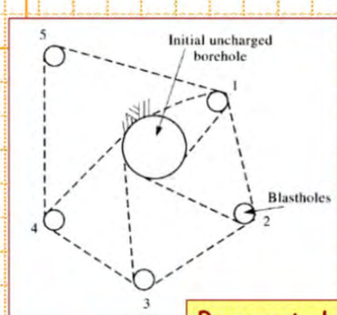
Burn cut

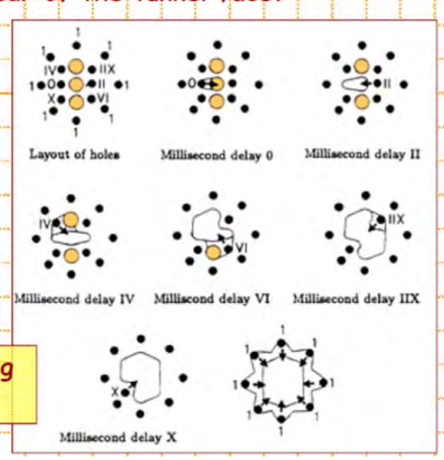


Arabic numerals: short delay periods
Roman numerals: half-second periods

Blasting Rounds – Burn Cut

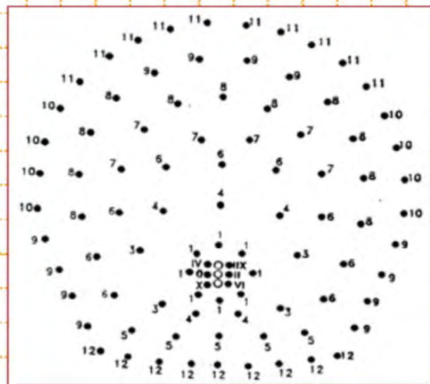
The correct design of a blast starts with the first hole to be detonated. In the case of a tunnel blast, the first requirement is to create a void into which rock broken by the blast can expand. This is generally achieved by a wedge or burn cut which is designed to create a clean void and to eject the rock originally contained in this void clear of the tunnel face.

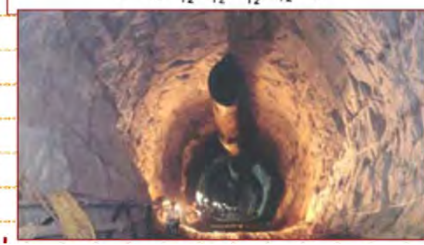


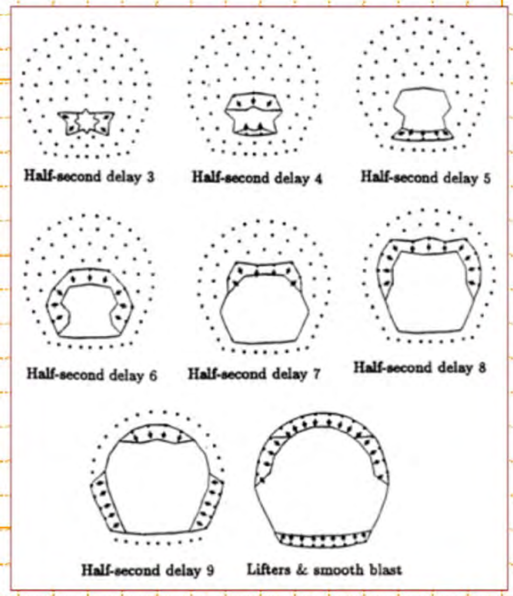


Burn cut designs using millisecond delays.

Blasting Rounds – Blast Pattern Design

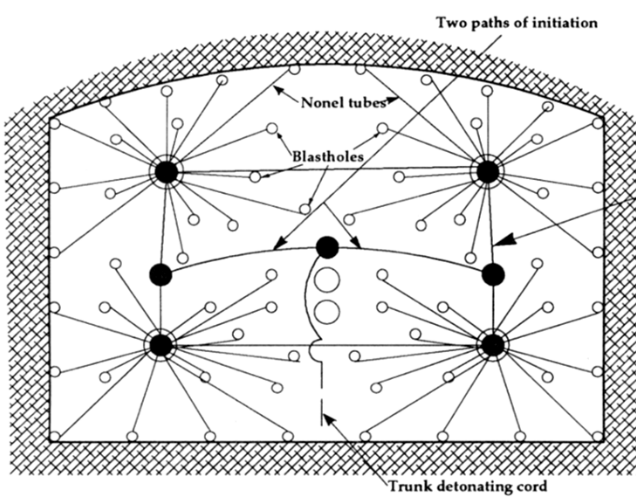




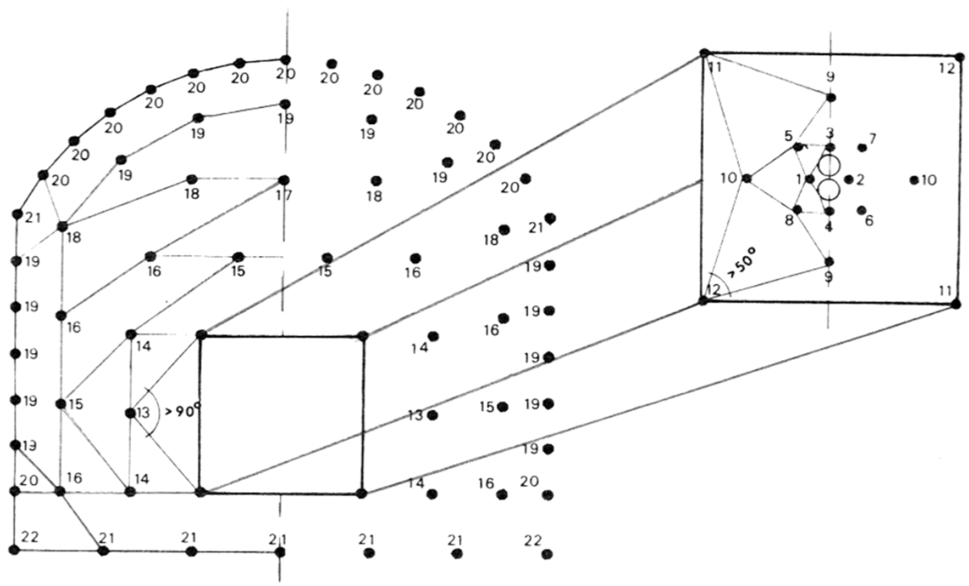


Sequence of detonation

For both fragmentation and throw, blasting efficiency depends on the delay sequence of blasthole detonation. Delayed detonation improves loadability of the entire cut, contributes to a better strata control and reduction of blast-induced vibrations.



Bunch blasting with Nonel tubes and detonating cord



Firing sequence for tunnel numerical order

Tunnel Blast Design

- Guiding Principles

The governing rule for underground blast design should be based on followings

- Progressive enlargement of free face
- Firing of holes towards maximum free face
- Planning of firing sequence to have maximum free face at the bottom of tunnel section so that maximum stoping can be done in favour of the gravity
- Creation of maximum free face before the contour holes are blasted so that minimum explosive quantity is required to break rock at the contour and the blast induced damage is restricted to a bare minimum

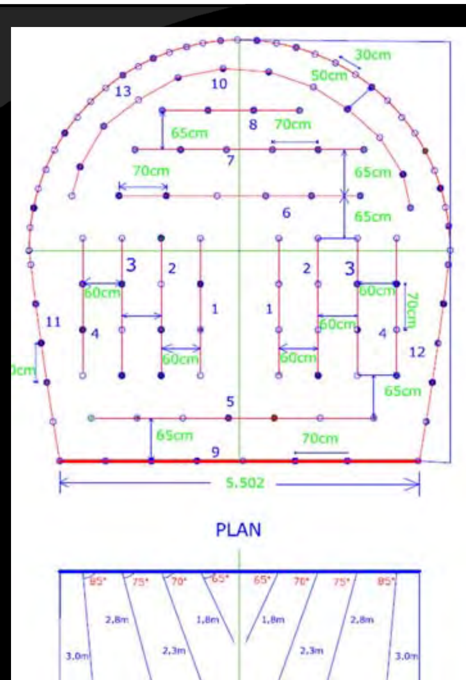
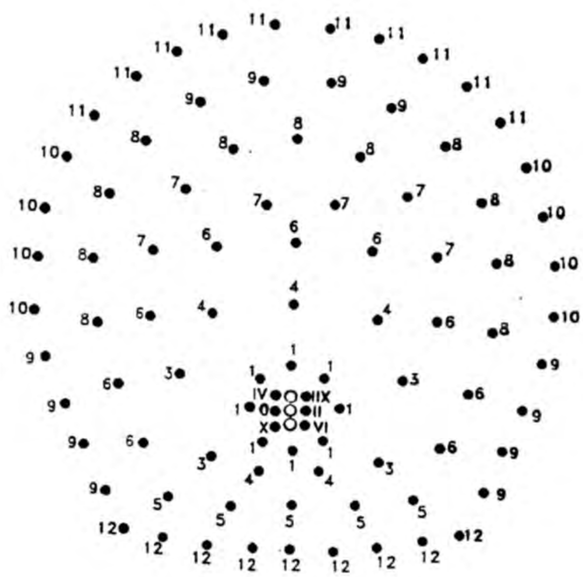
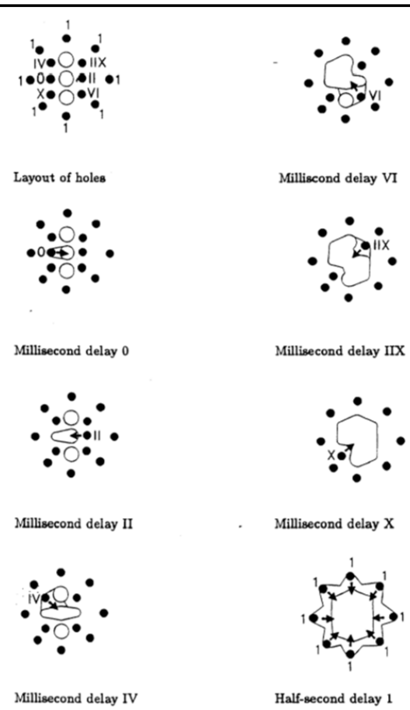


Fig:1 Blast design for tunnel excavation

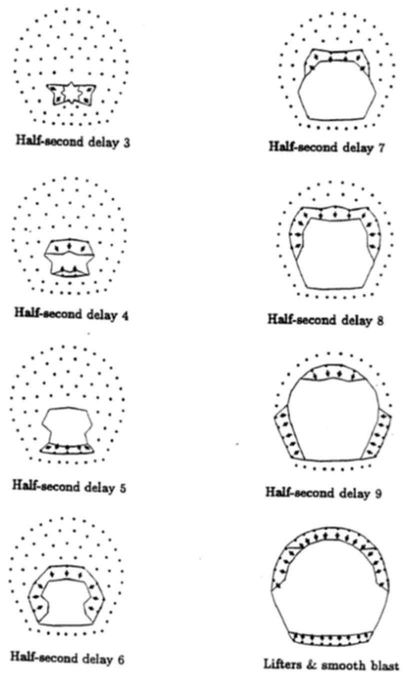


Holes	no	Dia mm	Explosives	Total wt. kg	Detonators
Burn	14	45	Gelamex 80, 18 sticks/hole	57	Millisec
Lifters	9	45	Gelamex 80, 16 sticks/hole	33	Half-sec
Perimeter	26	45	Gurit, 7 sticks/hole and Gelamex 80, 1 stick/hole	26	Half-sec
Others	44	45	Gelamex 80, 13 sticks/hole	130	Half-sec
Relief	3	75	No charge		
Total	96			246	

Blasthole pattern and charge details on the Victoria hydroelectric project in Sri Lanka. Roman numerals refer to the detonation sequence of millisecond delays in the burn cut, while Arabic numerals refer to the half-second delays in the remainder of the blast.



Development of a burn cut using millisecond delays.



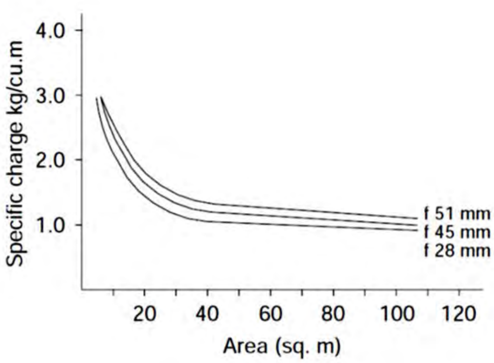
Use of half-second delays in the main blast and smooth blasting of the perimeter of a tunnel.

The correct design of a blast starts with the very first hole to be detonated. In the case of a tunnel blast, the first requirement is to create a void into which rock broken by the blast can expand. This is generally achieved by a wedge or burn cut which is designed to create a clean void and to eject the rock originally contained in this void clear of the tunnel face.

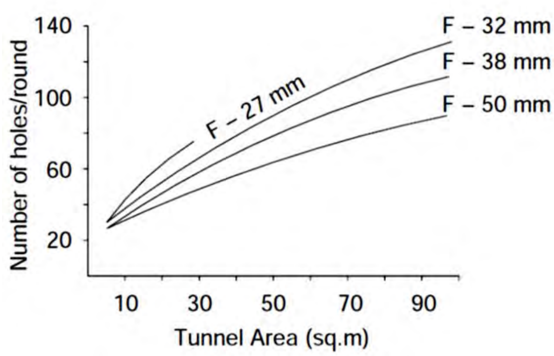
In today's drill and blast tunnelling in which multi-boom drilling machines are used, the most convenient method for creating the initial void is the burn cut. This involves drilling a pattern of carefully spaced parallel holes which are then charged with powerful explosive and detonated sequentially using millisecond delays.

Once a void has been created for the full length of the intended blast depth or `pull', the next step is to break the rock progressively into this void. This is generally achieved by sequentially detonating carefully spaced parallel holes, using one-half second delays. The purpose of using such long delays is to ensure that the rock broken by each successive blasthole has sufficient time to detach from the surrounding rock and to be ejected into the tunnel, leaving the necessary void into which the next blast will break.

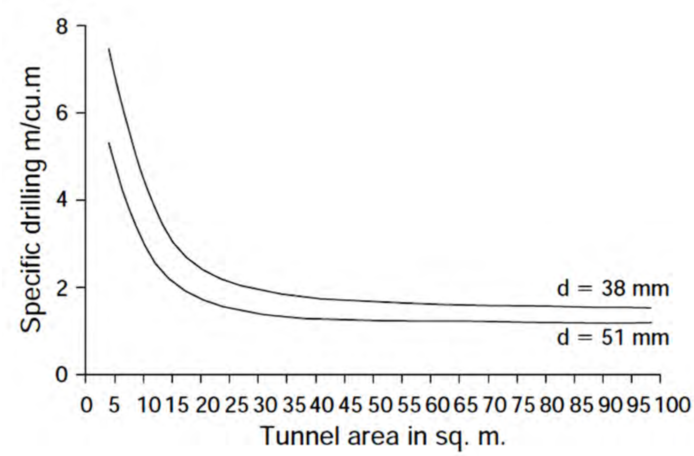
A final step is to use a smooth blast in which lightly charged perimeter holes are detonated simultaneously in order to peel off the remaining half to one metre of rock, leaving a clean excavation surface.



Influence of tunnel area on explosive charge required



Number of holes in relation to tunnel area



Influence of tunnel area on amount of drilling required

CHARGING AND BLASTING THE ROUNDS

Placement of primer:

Blastholes are usually charged with a continuous column of explosive cartridges. The effectiveness of blast in this practice is greatly affected by the location of the primer, and the stemming material and its length.

The primer can be placed at the blast hole collar – direct initiation, or at the blast hole bottom – inverse initiation. In the inverse initiation the blast energy is utilized to a great extent than in the direct initiation due to prolonged action of explosion product on the enclosing rock. The effect of inverse initiation increases with increase of hole depth.

CHARGING AND BLASTING THE ROUNDS

Stemming:

The amount of stemming material is specified by the safety regulations in some countries. In practice it is usually in the range of 0.6–0.8 m during drift/tunnel blasting.

Usually a mixture of clay and sand in the ratio of 1:4 is used. Water stemming (polyethylene tubes filled with water) contributes to adsorption of toxic gases and suppression of dust.

CHARGING AND BLASTING THE ROUNDS

Depth of round/hole:

There is no empirical formula as such to determine the depth of round but as a guideline, particularly in the workings of limited cross section, the *depth of hole =* $0.5\sqrt{S}$ *for angled cuts* and *for parallel hole cuts, it is equal to* $0.75\sqrt{S}$, (whereas, S is the cross sectional area of the drive).

CHARGING AND BLASTING THE ROUNDS

Charge density in cut-holes and rest of the face area:
Charge density i.e. explosive/m length in the cut-hole area is a function of diameter of empty hole, its distance from the charge hole and explosive density. However, since the distance between the cut-holes is so short and also the volume of rock likely to be blasted by the cluster of holes (i.e. in cut-hole area) is small, hence the required charge concentration/m length should be low. This means either explosives of low density; or higher density explosives with loose confinement, or with the use of some inert material like wood, as spacers, should be used. Except ammonium nitrate (AN) based explosives, all other commercial explosives are having higher density; that ranges 1.2–1.7 gms/c.c.

CHARGING AND BLASTING THE ROUNDS

Charge density in cut-holes and rest of the face area:
Hence, in the cut-holes the explosive cartridges must be just pushed in without any tight/hard tamping. Cut area after its blast creates a sufficient void/cavity. But the rest of the holes, which are drilled at an increased burden and spacing, and ultimately yield sufficient amount of rock, should be charged thoroughly with tight tamping. At the line holes (peripheral holes) the spacing between them should be reduced and so is the charge concentration by the use of either low density explosive or otherwise, to obtain smooth face profile with minimum over-break.

CHARGING AND BLASTING THE ROUNDS

Charge density in cut-holes and rest of the face area:
Selection of matching explosive for a particular hole diameter is vital; for example for a hole diameter in the range of 32–40 mm the explosive's VOD in the range of 2000–3000 m/sec is suitable. Similarly, large diameter holes require explosives of higher VOD. Cross sectional area influences explosive consumption considerably. For smaller tunnels higher powder factor is required and it gets reduced as their size increase.

SMOOTH BLASTING

In practice over-break along the planned shape, contour or configuration results after blasting a round. This result in higher costs on supports, more mucking, ore dilution, generation of cracks and exceptionally into roof falls. To have a better confirmation of the actual cross section with the designed one, a technique known as *smooth blasting*, is used.

In this technique extra care is taken while drilling and blasting the line (peripheral holes). Increasing the number of line holes and reducing spacing between them could achieve this. Practically, the spacing of lines holes for burden of 0.7–0.9 m is taken as 0.5 to 0.6 m. The line holes should be located as close to the working (tunnel walls) as the drill machine permits.

SMOOTH BLASTING

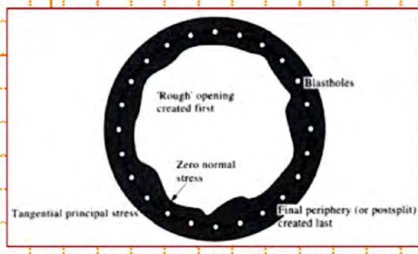
In smooth blasting, in the outer profile drilling must be increased to about 1.5 times the normal. Drilling quality by handhold machines is usually poor so far accuracy is concerned, whereas, drill jumbos can provide accurate drilling.

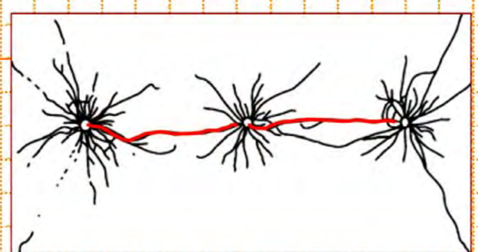
The importance of smooth blasting in drives and tunnels of large size is greatly realized than the workings of smaller cross-sections. Smooth blasting lowers the consumption of concrete for lining and promotes a wider use of shotcrete lining which reduces the roughness of tunnel surfaces; a desirable feature for ventilation point of view in mines; and water flow point of view in hydraulic tunnels.

Results achieved using well designed and carefully controlled blasting in a 19 foot diameter tunnel in gneiss in the Victoria hydroelectric project in Sri Lanka. Note that no support is required in this tunnel as a result of the minimal damage inflicted on the rock.

Specialized Blasting Techniques

During blasting, the explosive damage may not only occur according to the blasting round design, but there may also be extra rock damage behind the excavation boundary. To minimize damage to the rock, a smooth-wall blast may be used to create the final excavation surface.





Hudson & Harrison (1997)

The smooth-wall blast begins by creating a rough opening using a large bulk blast. This is followed by a smooth-wall blast along a series of closely spaced and lightly charged parallel holes, designed to create a fracture plane connecting the holes through by means of coalescing fractures.

Blasting Rounds – Fragmentation

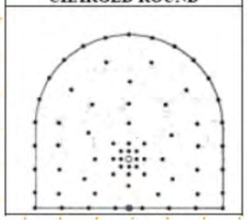
How efficiently muck from a working tunnel or surface excavation can be removed is a function of the blast fragmentation. Broken rock by volume is usually 50% greater than the *in situ* material. In mining, both the ore and waste has to be moved to surface for milling or disposal. Some waste material can be used underground to backfill mined voids. In tunnelling, everything has to be removed and dumped in fills – or if the material is right, may be removed and used for road ballast or concrete aggregate (which can sometimes then be re-used in the tunnel itself).

Blasting – Summary

CONTROLLABLE VARIABLES		
DRILLING	CHARGING	BLASTING
- Diameter drill hole - Drilled length - Drill pattern - Incorrect drilling	- Type of explosives - Energy of explosives - Charging method - Design of charging - Charged length - Firing pattern	- Firing system - Firing interval - Water (partly)

NON-CONTROLLABLE VARIABLES	
GEOLOGY	OTHER
- Rock parameters - Rock mass joint	- Incline/Decline - Water (partly)

CHARGED ROUND

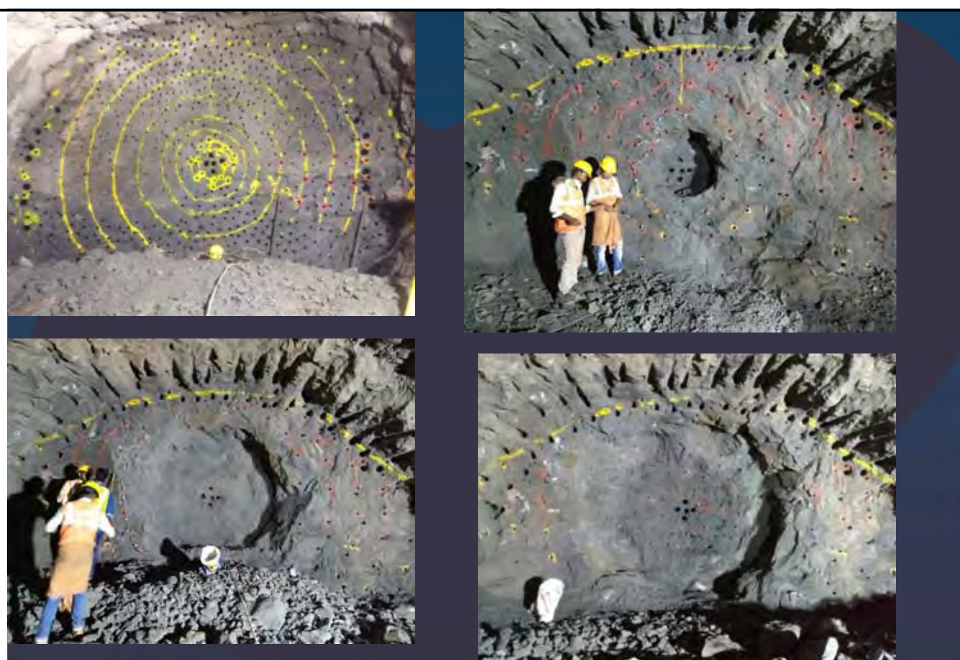


Typical production round fires in less than 7 seconds

RESULT

- Fragmentation
- Throw
- Muck pile shape
- Loadability
- Vibrations
- Advance per round
- Contour
- Flyrock
- Non-detonating holes
- Poor blast result

NTNU (1995)



Porgressive Enlargement

The Cavern for U/g Station

SHAFT SINKING

In mining, shafts form a system of vertically or inclined passageways, which are used for transportation of ore, refill, personnel, equipment, air, electricity, ventilation etc. In tunneling, shafts are used for transportation of equipment, air, electricity, ventilation etc.

An important requirement in shaft sinking is to provide optimum fragmentation of the rock so that it can be cleared quickly from the congested shaft-face area. Blasting operation is carried out against gravity, and the scatter of the broken rock is confined in the shaft. It is common to use generous distribution of explosives throughout the rock using a large number of small diameter (35 - 42 mm) shotholes.

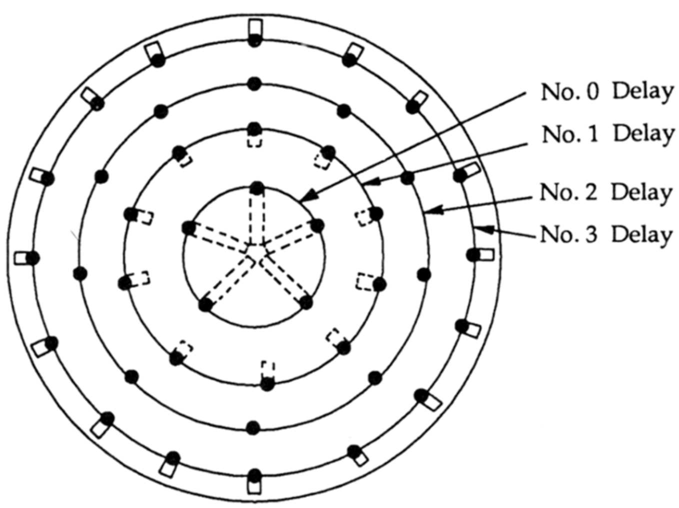
SHAFT SINKING

The number of holes N required for sinking a shaft of cross sectional area A in m^2 is given by:

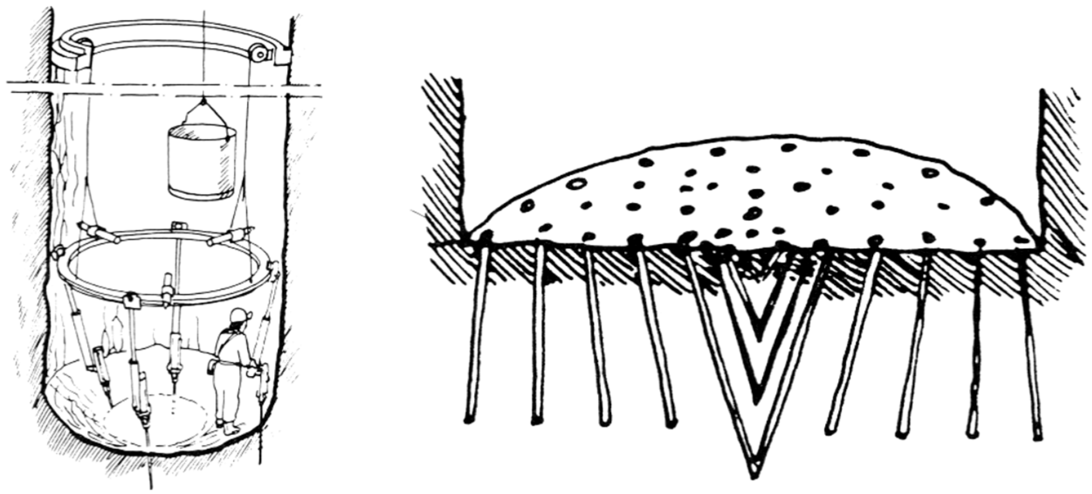
$$N = 2.5A + 22$$

The drilling patterns for shaft sinking are basically the same as those used in tunneling but generally the cone cut is favoured.

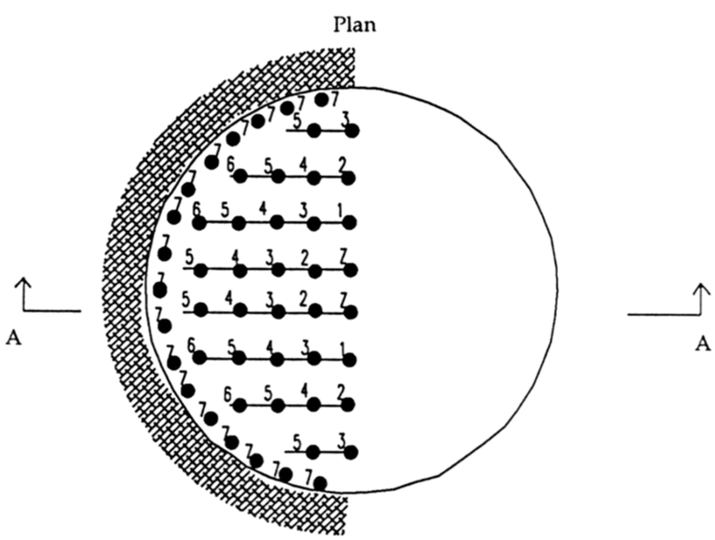
Another commonly used pattern, unique to shaft sinking, is the bench cut.



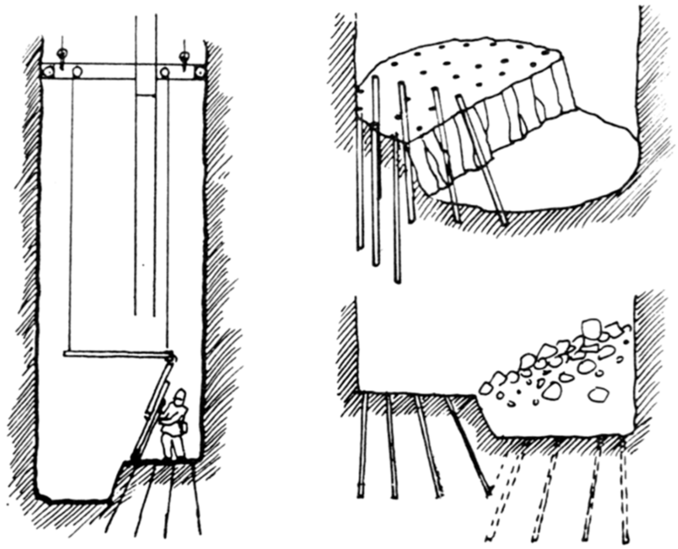
Full-face blasthole pattern for shaft sinking



Shaft sinking with cone or pyramid cut



A bench-cut blasthole for shaft sinking



Shaft sinking by benching

SHAFT SINKING

The explosives used in shaft sinking must always be water resistant. Even if the ground is dry, the flushing water from the drilling machines will always stay in the blastholes.

Suitable explosives such as Emulex 150 or any suitable NG based explosives are easily tamped to utilize the hole volume well.

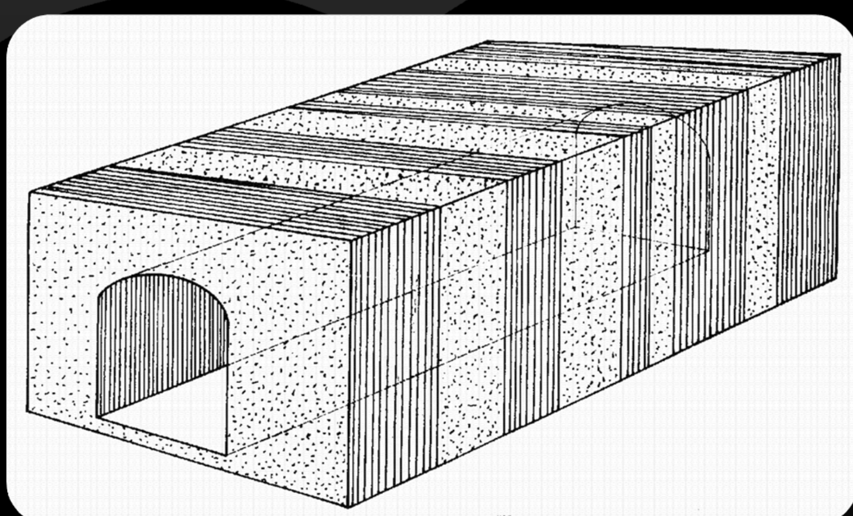
The powder factor in shaft sinking is rather high, ranging from 2.0 kg/m^3 to 4.0 kg/m^3.

SHAFT SINKING

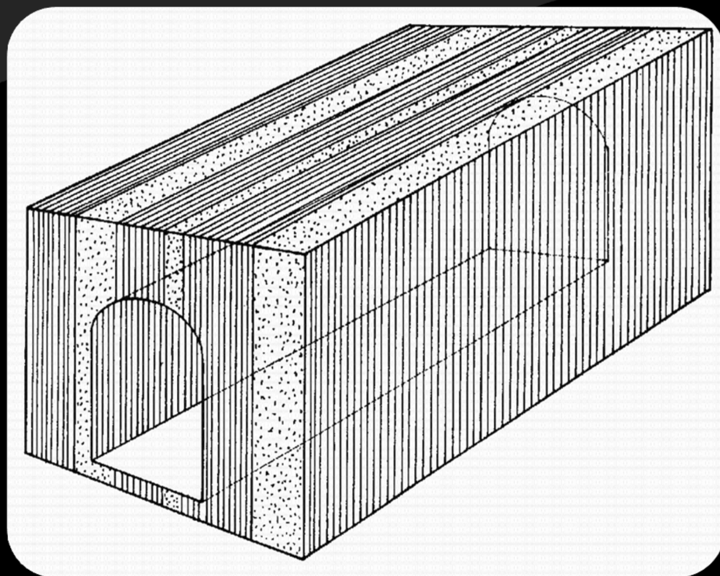
If charges are fired electrically, great care must be taken in wiring the circuit. Since more than 100 detonators can be involved in each blast, they are connected either in parallel or in series-parallel.

It is therefore, important to ensure that the resistance of the circuit is properly balanced and that no charged hole is omitted from the circuit. Due to higher risk of such errors in the generally unfavourable shaft sinking environment, Nonel type detonators are increasingly preferred for initiation.

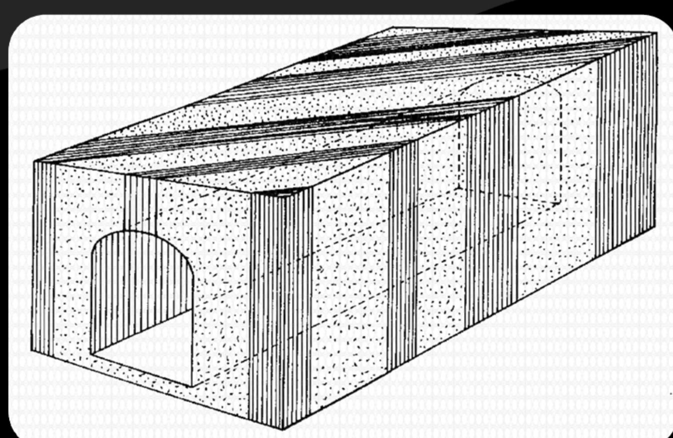
Effects of joint orientations on blast performance



Joints normal to tunnel direction favourable for good pull

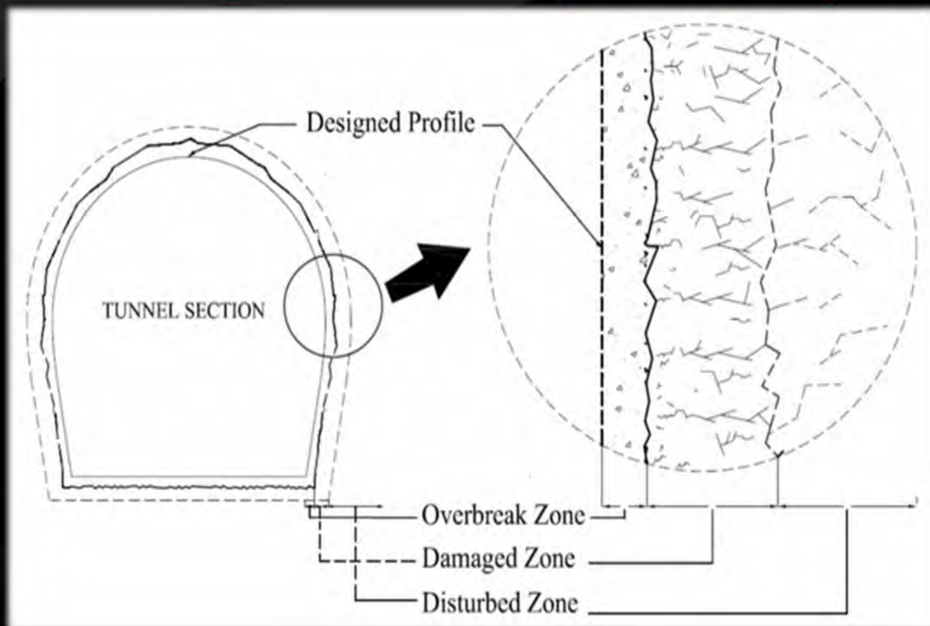


Poor advance per round predicted with joints striking parallel to tunnel advance direction

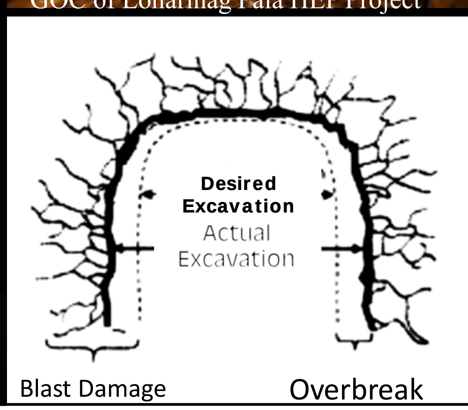


Right side wall more prone to breakage with joints at angle to tunnel direction

Damage around an Underground Opening



Method of Excavation	Damage Level	Reduction in RMR (%)
Mechanical Excavation	Slight Damage	0 - 6
Controlled Blasting Techniques	Moderate Damage	6 -10
Poor Conventional Blasting	Severe Damage	10 - 20



- Blasting inherently induces damage to the rock mass
- Overbreak upto 10-15% is very common
- Studies have shown that the damage extends much beyond the overbreak zone
- 10% OB means constructing additional 100 m of tunnel for every 1.0 km of planned tunnel length

Damage zones around an underground opening

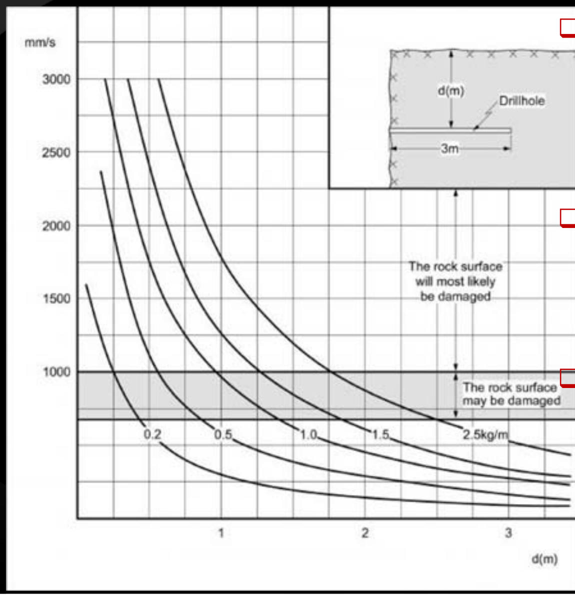
- **Overbreak Zone**
  - This zone represents difference in "as designed" and "as excavated".
  - In this zone rocks pieces and slabs detach completely from the rock mass.
  - Overbreak zone (OBZ) is measured in per cent of extra volume of rock.
  - OB (%) vary from as 5 % to 25 %. 10% OB is very common in civil tunnels
  - OB significantly increase cost of the project
- **Damaged Zone**
  - Zone around overbreak or failed zone where irreversible changes in the rock mass properties takes places
  - Reduction in mechanical properties due to creation of micro-cracks and fractures.
  - Determined as a distance upto which there is 10% reduction in P-wave velocity
- **Disturbed Zone**
  - Zone beyond damaged zone where insignificant and reversible changes takes place
  - In civil tunnels, disturbed zone have insignificant importance but play vital role in nuclear repositories as it affects flowpath of contaminants
  - It does not affcet safety and productivity of civil tunnel

Factors affecting blast damage zone

- Rock Mass Properties
  - Strength
  - Discontinuities and it's properties
  - In-situ stress
- Explosive Properties
  - Strength, diameter, cartridges shape and size etc.
- Blast Design parameters
  - Drilling pattern, Maximum charge per delay, Specific charge, perimeter charge factor, initiation and firing sequence

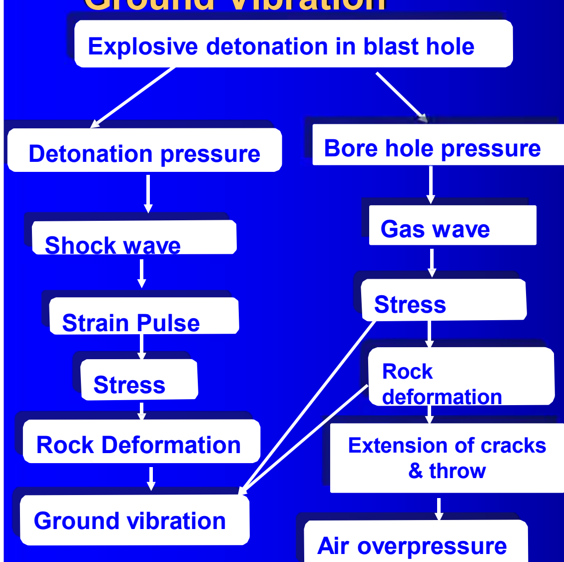
Damage Produced in Rock Mass due to different Charge Density in Contour Hole and associated PPV values in a Tunnel blasting

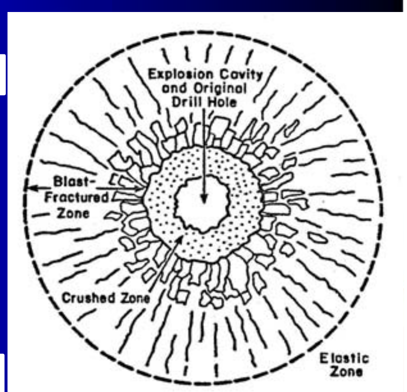
(Holmberg and Persson (1979))



- To control over-breakage, it is not only essential to use small charge but also necessary to have greater dispersion of charge
- Damage zone around a blast hole for a charge concentration of 0.2 kg/m is only 0.5 m, whereas for 2.5 kg/m it is 2.8 m
- With same quantity of explosive a blast with large number of small holes will produce a more smooth and sound surface than that with a small number of holes with big charges

Ground Vibration





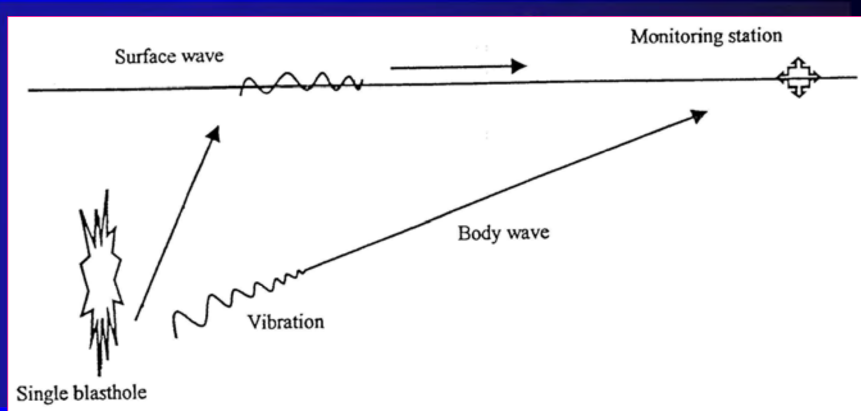
Fracturing and deformation around an explosion in rock

PROCESS OF GENERATION OF GROUND VIBRATIONS & AIR OVERPRESSURE

Problems due to Blast Induced Damages

- Loosening, dislocation and disturbance of rock
- Overbreak
- Roof fall
- Reduction in strength of rock mass
- Unstable underground openings
- Extra cost of support/ backfill/ reinforcement
- Increase in production time and cycle time
- Loss of property (men and machinery)

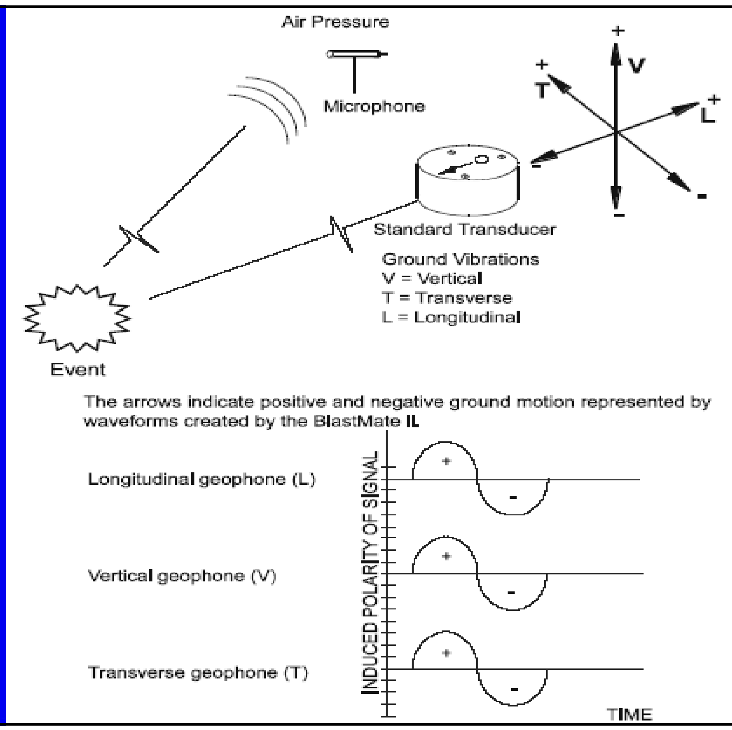
Vibration



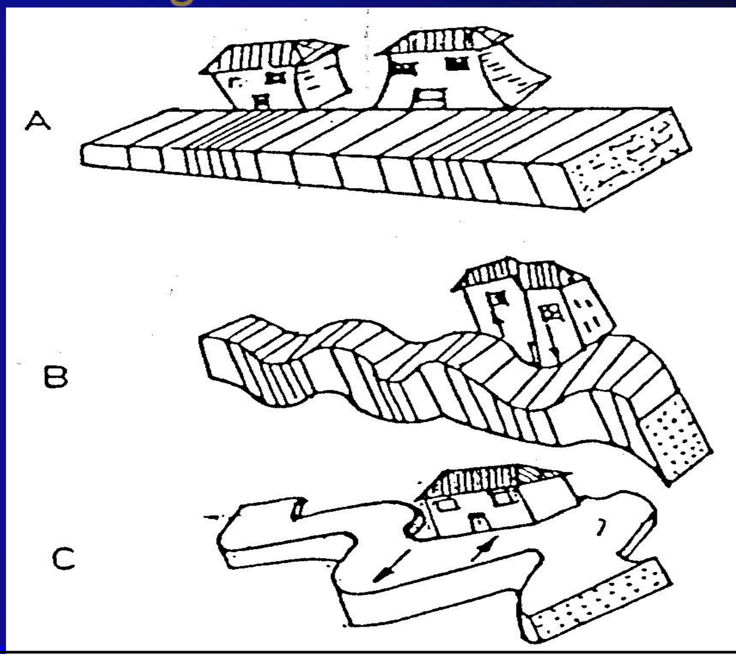
Parameters influencing ground vibrations (Wiss and Linehan, 1978).

Parameters	Influence on PPV		
	Significant	Moderately significant	Insignificant
Controllable variables			
Charge weight per delay	X		
Length of delay	X		
Burden and spacing		X	
Stemming amount			X
Type of stemming			X
Charge length and diameter			X
Angle of borehole			X
Direction of initiation		X	
Charge weight per blast			X
Charge depth			X
Bare Vs. covered detonating cord			
Uncontrollable variables			
General surface terrain			
Type and depth of overburden (Confinement)		X	
Wind			x

Vibration Monitoring



Damage to structures



IMPORTANT PARAMETERS OF GROUND VIBRATION

- PROPAGATION VELOCITY
- PARTICLE DISPLACEMENT
- PARTICLE VELOCITY
- PARTICLE ACCELERATION
- FREQUENCY
- DURATION

PPV AND STRUCTURAL DAMAGE

- Peak particle displacement, velocity and acceleration along with their associated frequencies are three possibilities to correlate with damage.
- Damage is more closely associated with peak particle velocity (Duvall & Fogelson, 1962)
- Physical basis lies in the following simple relationship

$$\varepsilon = V_{resp}/V_C \times 10^{-6}$$

where V_{resp} is the peak response velocity (mm/s) and V_C is the wave propagation velocity (km/s).

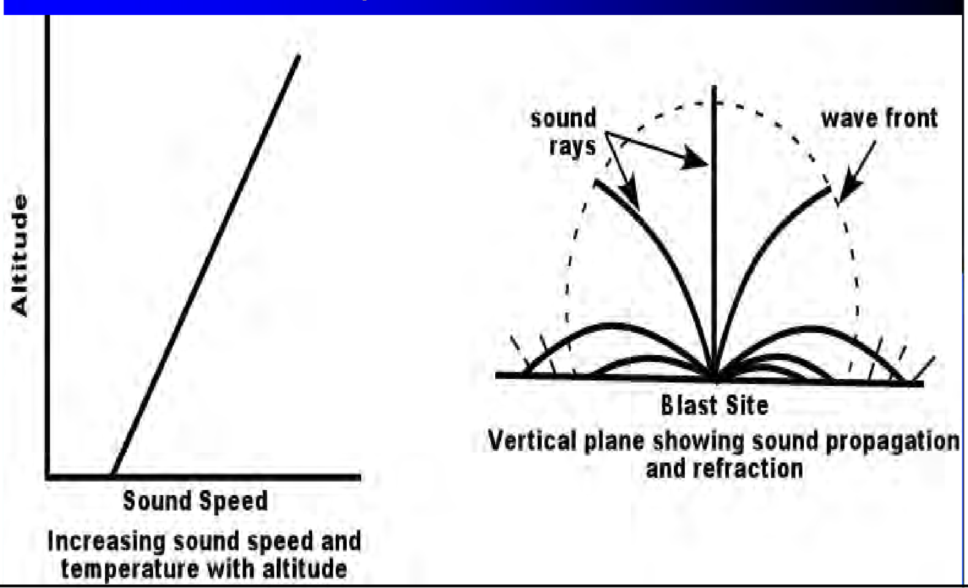
Safety criteria are most commonly defined in terms of PPV and associated frequency.

Minimate - Vibration Monitoring Equipment

Permissible Limit of Vibration (PPV, mm/s)

	<8 Hz	8-25 Hz	> 25 Hz
(A) Buildings /structures not belonging to the owner			
Domestic houses / structures (Kuchcha, brick & cement)	**5**	**10**	**15**
Industrial buildings	**10**	**20**	**25**
Objects of historical importance and sensitive structures	**2**	**5**	**10**
Domestic houses / structures	10	15	25
Industrial buildings	15	25	50

Effect of Atmospheric Conditions – Temperature Inversion



Air overpressure limit

Air Overpressure limits	Damage potential
180 dB	Some structural Damage
171 dB	General window breakages
140 dB	Occasional Window breakage
134 dB	US Bureau of mines recommendations for large scale surface mine blasting.

BLASTING VIBRATION LIMITS FOR MASS CONCRETE

Concrete Age from Batching	Allowable Particle Velocity ips (cm/s) x Distance Factor (DF)
0 to 4 hours	4 (10.2) x DF
4 hours to 1 day	6 (15.2) x DF
1 to 3 days	9 (22.9) x DF
3 to 7 days	12 (30.5) x DF
7 to 10 days	15 (38.1) x DF
10 days or more	20 (50.8) x DF

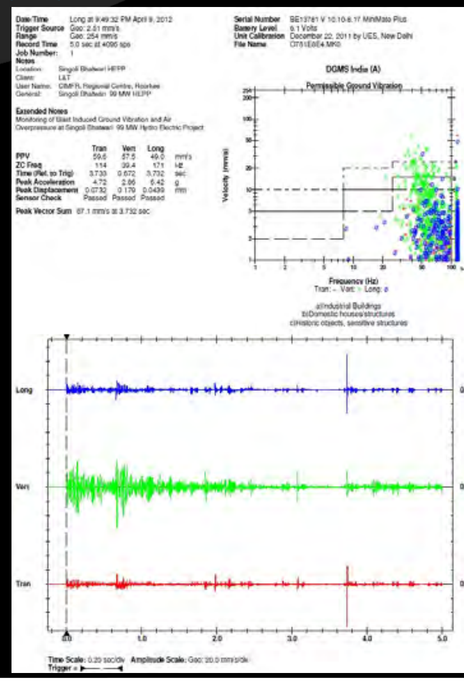
The Distance Factor is defined as follows:

Distance = 0 - 50 ft (0 – 15 m) DF = 1.0
Distance = 50 - 150 ft (15 – 46 m) DF = 0.8
Distance = 150 - 250 ft (46 – 76 m) DF = 0.7
Distance = Over 250 ft (76 m) DF = 0.6

Vibration Attenuation Characteristics

Parameters obtained from Vibration Monitoring

- Peak particle velocity, mm/s
- Peak vector sum, mm/s
- Frequency, Hz
- Acceleration, g, and
- Displacement, mm
- Sensor location and distance
- maximum charge per delay



Blast Design Parameters

- Total charge (T)
- Maximum charge per Delay (W)
- Specific Charge (q)
- Perimeter charge factor (q_p)
- Hole depth (d), m
- Face Advancement (l), m
- Initiation and Firing sequence

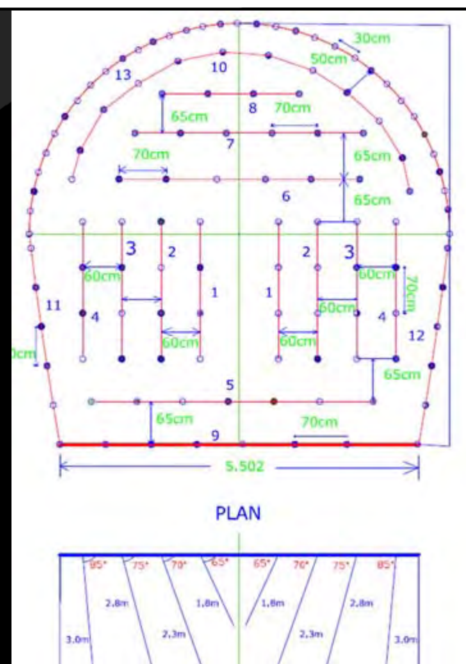


Fig:1 Blast design for tunnel excavation